

Multi-agentic implementation of a 3D topology optimization framework for compliance and linearized buckling via weighted sum method

Gabriele Ricatto

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Abstract

This paper presents a rigorous theoretical and computational framework for three-dimensional (3D) multi-objective continuum topology optimization, focusing on the trade-off between structural compliance (global stiffness) and linear buckling stability. A custom Finite Element Method (FEM) solver based on 8-node trilinear hexahedral isoparametric elements (H8) applied to a structured Cartesian voxel grid is implemented. Material distribution is parameterized using the Solid Isotropic Material with Penalization (SIMP) approach, grounded in the 3D Hashin-Shtrikman composite bounds ($p_K = 3$).

To explore the Pareto-optimal frontier between global stiffness and the fundamental Buckling Load Factor (BLF), a weighted sum method with dynamic L_1 -norm gradient normalization is developed, guaranteeing Pareto-proper efficiency under Geoffrion’s theorem. To eliminate runtime numerical quadrature during the optimization loop, the 3D geometric stiffness matrix is synthesized by decomposing the Cauchy stress tensor into six constant invariant 24×24 basis matrices, accelerating assembly by two orders of magnitude while suppressing spurious void buckling modes through differential penalization ($p_G \geq p_K$). Exact analytical sensitivities are derived via the adjoint state method requiring a single back-substitution. The numerical architecture is optimized for execution on commodity hardware with 8GB RAM (AMD Ryzen 7 4700U), consuming under 100 MB peak RAM for standard test domains, and seamlessly interfaces with a watertight STL surface extraction pipeline.

1 Introduction

In conventional minimum compliance topology optimization governed purely by linear elasticity, applying an external force $\mathbf{F}$ or its exact inverse $-\mathbf{F}$ produces an identical optimized topology. Static equilibrium is governed by Hooke’s law:

$$\mathbf{K}(\rho)\mathbf{U} = \mathbf{F} \quad (1)$$

where $\mathbf{K}$ is the symmetric, positive-definite global stiffness matrix, $\mathbf{U}$ is the nodal displacement vector, and $\mathbf{F}$ is the applied external force vector. Inverting the load vector to $-\mathbf{F}$ yields an identically reversed displacement field:

$$\mathbf{U}_{-\mathbf{F}} = \mathbf{K}^{-1}(-\mathbf{F}) = -\mathbf{U} \quad (2)$$

Structural compliance C represents the total external work done by the loads:

$$C = \mathbf{F}^T \mathbf{U} = \mathbf{U}^T \mathbf{K} \mathbf{U} = \sum_{e=1}^m E(\rho_e) \mathbf{u}_e^T \mathbf{k}_0 \mathbf{u}_e \quad (3)$$

Evaluating the compliance under the reversed load $-\mathbf{F}$ gives:

$$C_{-\mathbf{F}} = (-\mathbf{F})^T (-\mathbf{U}) = \mathbf{F}^T \mathbf{U} = C \quad (4)$$

Because the strain energy is a quadratic form in the nodal displacements, it remains strictly positive and insensitive to the sign of the displacement vector. Under small-strain linear elasticity and symmetric material behavior ($E_{\text{tension}} = E_{\text{compression}}$), tensile and compressive stresses contribute identically to structural compliance.

Consequently, a compliance-only optimizer cannot differentiate between a slender tension tie and an Euler column subjected to axial compression. Under compression, slender members undergo catastrophic geometric instability (buckling) at loads orders of magnitude below their yield limit. Breaking this artificial symmetry requires the explicit inclusion of Linearized Buckling Analysis (LBA) into the optimization objective [1, 7].

2 FEM Implementation: 3D Isoparametric H8 Element

For the implementation of the 3D FEM solver, we rely on eight-node trilinear hexahedral isoparametric elements (H8) applied to a fully structured Cartesian voxel grid. The rectangular elements have dimensions $dx \times dy \times dz$ and volume $V_e = dx \cdot dy \cdot dz$, aligned with the global reference system, with the origin of the natural coordinates $(\xi, \eta, \zeta) \in [-1, 1]^3$ centered in the centroid of each element (x_c, y_c, z_c) :

$$x = x_c + \frac{dx}{2}\xi, \quad y = y_c + \frac{dy}{2}\eta, \quad z = z_c + \frac{dz}{2}\zeta \quad (5)$$

with $\xi, \eta, \zeta \in [-1, 1]$. The Jacobian matrix $[\mathbf{J}]$ is strictly diagonal:

$$[\mathbf{J}] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \\ \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta} \end{bmatrix} = \begin{bmatrix} dx/2 & 0 & 0 \\ 0 & dy/2 & 0 \\ 0 & 0 & dz/2 \end{bmatrix} \implies \det(\mathbf{J}) = \frac{1}{8}dx \, dy \, dz = \frac{1}{8}V_e \quad (6)$$

and its inverse is:

$$[\mathbf{J}]^{-1} = \begin{bmatrix} 2/dx & 0 & 0 \\ 0 & 2/dy & 0 \\ 0 & 0 & 2/dz \end{bmatrix} \quad (7)$$

The trilinear shape functions $N_i(\xi, \eta, \zeta)$ for the 8 local nodes are:

$$N_i(\xi, \eta, \zeta) = \frac{1}{8}(1 + \xi_i\xi)(1 + \eta_i\eta)(1 + \zeta_i\zeta), \quad i \in \{1, 2, \dots, 8\} \quad (8)$$

where $(\xi_i, \eta_i, \zeta_i) \in \{-1, 1\}^3$ denote the corner coordinates in natural space.

2.1 Kinematic Degrees of Freedom and B-Matrix

Each node possesses 3 translational degrees of freedom (u, v, w) , yielding $8 \times 3 = 24$ DOFs per element. The element displacement vector is:

$$\mathbf{u}_e = [u_1 \quad v_1 \quad w_1 \quad u_2 \quad v_2 \quad w_2 \quad \dots \quad u_8 \quad v_8 \quad w_8]^T \in \mathbb{R}^{24} \quad (9)$$

The engineering strain vector $\boldsymbol{\varepsilon} = [\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}, \gamma_{yz}, \gamma_{zx}, \gamma_{xy}]^T \in \mathbb{R}^6$ is interpolated via the 6×24 strain-displacement matrix $\mathbf{B}$:

$$\boldsymbol{\varepsilon}(\xi, \eta, \zeta) = \mathbf{B}(\xi, \eta, \zeta)\mathbf{u}_e = [\mathbf{B}_1 \quad \mathbf{B}_2 \quad \dots \quad \mathbf{B}_8] \mathbf{u}_e \quad (10)$$

where each 6×3 nodal submatrix $\mathbf{B}_i$ is defined as:

$$\mathbf{B}_i = \begin{bmatrix} \frac{2}{dx} \frac{\partial N_i}{\partial \xi} & 0 & 0 \\ 0 & \frac{2}{dy} \frac{\partial N_i}{\partial \eta} & 0 \\ 0 & 0 & \frac{2}{dz} \frac{\partial N_i}{\partial \zeta} \\ 0 & \frac{2}{dz} \frac{\partial N_i}{\partial \zeta} & \frac{2}{dy} \frac{\partial N_i}{\partial \eta} \\ \frac{2}{dz} \frac{\partial N_i}{\partial \zeta} & 0 & \frac{2}{dx} \frac{\partial N_i}{\partial \xi} \\ \frac{2}{dy} \frac{\partial N_i}{\partial \eta} & \frac{2}{dx} \frac{\partial N_i}{\partial \xi} & 0 \end{bmatrix} \quad (11)$$

2.2 Element Stiffness Matrix $\mathbf{k}_0$ (24×24)

For isotropic 3D linear elasticity characterized by Young's modulus E_0 and Poisson's ratio ν , the constitutive matrix $\mathbf{D} \in \mathbb{R}^{6 \times 6}$ is:

$$\mathbf{D} = \frac{E_0}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \quad (12)$$

The base element stiffness matrix $\mathbf{k}_0 \in \mathbb{R}^{24 \times 24}$ is evaluated via exact $2 \times 2 \times 2$ Gauss-Legendre quadrature:

$$\mathbf{k}_0 = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 \mathbf{B}^T \mathbf{D} \mathbf{B} \det(\mathbf{J}) d\xi d\eta d\zeta \quad (13)$$

Spectral analysis confirms that $\mathbf{k}_0$ possesses **exactly 6 zero eigenvalues** (three rigid translations and three rigid rotations) and **18 strictly positive eigenvalues** corresponding to independent elastic strain energy modes, verifying freedom from hourglassing instabilities.

2.3 3D Geometric Stiffness Decomposition into 6 Invariant Basis Matrices

The elemental geometric stiffness matrix $\mathbf{G}_0^e$ is derived from the linearization of the Principle of Virtual Work under finite Green-Lagrange strain:

$$\mathbf{G}_0^e = \int_{\Omega_e} \mathbf{B}_{nl}^T \mathbf{M}_\sigma \mathbf{B}_{nl} d\Omega \quad (14)$$

where $\mathbf{B}_{nl} \in \mathbb{R}^{9 \times 24}$ contains the spatial gradients of the displacement components, and $\mathbf{M}_\sigma = \mathbf{I}_3 \otimes \boldsymbol{\sigma}_e \in \mathbb{R}^{9 \times 9}$ embeds the 3×3 Cauchy stress tensor:

$$\boldsymbol{\sigma}_e = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{bmatrix} \quad (15)$$

Decomposing $\mathbf{M}_\sigma$ into its 6 independent scalar stress components [7]:

$$\mathbf{M}_\sigma = \sigma_{xx} \mathbf{T}_{xx} + \sigma_{yy} \mathbf{T}_{yy} + \sigma_{zz} \mathbf{T}_{zz} + \tau_{yz} \mathbf{T}_{yz} + \tau_{xz} \mathbf{T}_{xz} + \tau_{xy} \mathbf{T}_{xy} \quad (16)$$

where $\mathbf{T}_k = \mathbf{I}_3 \otimes \mathbf{t}_k$ are constant binary basis matrices. Substituting into the integral:

$$\mathbf{G}_0^e = \sum_{k \in \{xx, yy, zz, yz, xz, xy\}} \sigma_{k,e} \mathbf{G}_{0,k} \quad (17)$$

where the six constant 24×24 geometric stiffness basis matrices are defined as:

$$\mathbf{G}_{0,k} = \int_{\Omega_e} \mathbf{B}_{nl}^T \mathbf{T}_k \mathbf{B}_{nl} d\Omega = V_e \cdot \mathbf{B}_{nl}(0,0,0)^T \mathbf{T}_k \mathbf{B}_{nl}(0,0,0) \quad (18)$$

Computational Advantage: Because all six basis matrices $\mathbf{G}_{0,k}$ depend solely on element dimensions (dx, dy, dz) , they are precomputed **exactly once** during initialization. Inside the optimization iterations, element geometric stiffness is formed by simple scalar multiplication, accelerating assembly by more than two orders of magnitude.

3 Principal Stress Decomposition: Signed Stress Metric for Tension and Compression

At each hexahedral element e , the centroidal engineering strain $\boldsymbol{\varepsilon}_e = \mathbf{B}(0,0,0)\mathbf{u}_e$ yields the elemental Cauchy stress tensor $\boldsymbol{\sigma}_e \in \mathbb{R}^{3 \times 3}$:

$$\boldsymbol{\sigma}_e = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{bmatrix} \quad (19)$$

Because $\boldsymbol{\sigma}_e$ is real and symmetric ($\boldsymbol{\sigma}_e = \boldsymbol{\sigma}_e^T$), the spectral theorem guarantees the existence of three real eigenvalues and a mutually orthogonal triad of principal unit vectors $\{\mathbf{n}_I, \mathbf{n}_{II}, \mathbf{n}_{III}\}$. The stress tensor admits the spectral decomposition:

$$\boldsymbol{\sigma}_e = \sum_{p \in \{I, II, III\}} \sigma_p \mathbf{n}_p \otimes \mathbf{n}_p = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^T \quad (20)$$

where $\mathbf{Q} = [\mathbf{n}_I, \mathbf{n}_{II}, \mathbf{n}_{III}] \in \text{SO}(3)$ is the orthogonal transformation matrix, and the diagonal matrix is $\boldsymbol{\Lambda} = \text{diag}(\sigma_I, \sigma_{II}, \sigma_{III})$. The principal stresses are algebraically ordered:

$$\sigma_I \geq \sigma_{II} \geq \sigma_{III} \quad (21)$$

where σ_I denotes the maximum principal tensile stress, σ_{III} denotes the maximum compressive stress (the most negative eigenvalue), and σ_{II} represents the intermediate principal stress.

In classical structural analysis, scalar stress invariants such as the von Mises equivalent stress σ_{vM} :

$$\sigma_{vM} = \sqrt{\frac{1}{2} [(\sigma_I - \sigma_{II})^2 + (\sigma_{II} - \sigma_{III})^2 + (\sigma_{III} - \sigma_I)^2]} \geq 0 \quad (22)$$

are strictly non-negative and blind to hydrostatic sign reversals. As a result, σ_{vM} cannot distinguish between an intense tensile state and a severe compressive state.

To resolve this limitation and distinguish tensile load paths from stability-critical members, we define the *Signed Principal Stress* metric σ_{signed} :

$$\sigma_{\text{signed}} = \begin{cases} \sigma_I & \text{if } |\sigma_I| \geq |\sigma_{III}| \\ \sigma_{III} & \text{if } |\sigma_{III}| > |\sigma_I| \end{cases} \quad (23)$$

This scalar field isolates the dominant principal stress magnitude while preserving its physical sign:

- **Tensile Chords** ($\sigma_{\text{signed}} > 0$): Elements where $\sigma_{\text{signed}} > 0$ reside in a tensile stress regime governed by σ_I . Tensile forces induce positive-definite geometric stiffness contributions (stress stiffening), rendering tensile ties inherently stable against geometric bifurcation. Under combined compliance and stability optimization, these members can be realized as highly slender, direct tensile ties without incurring buckling penalties.

- **Euler Buckling-Prone Compressive Chords** ($\sigma_{\text{signed}} < 0$): Elements where $\sigma_{\text{signed}} < 0$ are dominated by compressive stress ($\sigma_{III} < 0$). In accordance with Euler’s critical buckling load formulation:

$$P_{\text{cr}} = \frac{\pi^2 EI}{(KL)^2} \quad (24)$$

slender structural members subjected to compressive axial loads are prone to catastrophic elastic bifurcation instabilities at stress levels well below the material yield limit. The compressive eigenvalues in $\boldsymbol{\sigma}_e$ generate negative contributions to the geometric stiffness matrix $\mathbf{G}$, driving the structure toward geometric collapse ($\det(\mathbf{K} + \lambda \mathbf{G}) \rightarrow 0$).

By monitoring σ_{signed} across the design domain, the topology optimization framework explicitly identifies compressive struts that require larger second moments of area I —motivating the formation of lateral bracing, webbed cross-sections, or thickened compressive arches—while permitting pure tension ties to remain slender.

4 Optimization Problem Setup

The spatial domain is discretized into m equal elements. The pseudo-density field is defined as $\hat{\mathbf{x}} = \{\hat{x}_e\}_{e=1}^m \in [0, 1]^m$. Intermediate densities are filtered through a 3D spherical cone filter ($r_{\min}$) [2, 7]:

$$\tilde{x}_e = \frac{\sum_{i=1}^m H_{ei} x_i}{\sum_{i=1}^m H_{ei}}, \quad H_{ei} = \max(0, r_{\min} - \text{dist}(\Omega_e, \Omega_i)) \quad (25)$$

and projected toward discrete 0/1 boundaries via smoothed Heaviside projection [8]:

$$\hat{x}_e = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{x}_e - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \quad (26)$$

with threshold $\eta = 0.5$ and continuation parameter $\beta \in [1, 32]$.

4.1 SIMP Interpolation and Differential Penalization

The elastic and geometric stiffnesses are parameterized using differential penalization [1, 7]:

$$E_K(\hat{x}_e) = E_{\min} + (E_0 - E_{\min}) \hat{x}_e^{p_K} \quad (27)$$

$$E_G(\hat{x}_e) = E_0 \hat{x}_e^{p_G} \quad (28)$$

with $p_K = 3.0$ (grounded in 3D Hashin-Shtrikman bounds) and $p_G = 3.0$ (or 6.0). In void regions ($\hat{x}_e \rightarrow 0$):

$$\frac{E_G(\hat{x}_e)}{E_K(\hat{x}_e)} \xrightarrow{\hat{x}_e \rightarrow 0} \frac{0}{E_{\min}} = 0 \quad (29)$$

This completely suppresses spurious void buckling modes in empty space without corrupting the physical low-frequency structural spectrum.

4.2 Linearized Buckling Eigenvalue Problem

Linearized structural stability is governed by the generalized eigenvalue problem:

$$(\mathbf{K} + \lambda_j \mathbf{G}) \phi_j = \mathbf{0} \quad (30)$$

To prevent singularities in void elements, the reciprocal eigenvalue $\mu_j = 1/\lambda_j$ is evaluated [1, 7]:

$$(\mathbf{G} + \mu_j \mathbf{K}) \phi_j = \mathbf{0} \quad (31)$$

where the fundamental critical buckling load factor is $\lambda_1 = 1/\mu_1$ with $\mu_1 = \max_j(\mu_j)$.

4.3 Multi-Objective Formulation and Dynamic L_1 Sensitivity Scaling

To ensure global static load bearing while simultaneously stiffening compressive members against buckling, we formulate the multi-objective optimization problem:

$$\min_{\mathbf{x}} \{C(\hat{\mathbf{x}}), -\mu_1(\hat{\mathbf{x}})\} \quad \text{subject to} \quad \frac{1}{m} \sum_{e=1}^m \hat{x}_e \leq V_f \quad (32)$$

Because these two objectives are competing, the solution space forms a Pareto-optimal frontier. We scalarize the problem into a combined objective function $F(\hat{\mathbf{x}})$ using the weighted sum method [3]:

$$F(\hat{\mathbf{x}}) = (1 - \alpha)C + \alpha\mu_1, \quad \alpha \in [0, 1] \quad (33)$$

Because compliance and eigenvalue sensitivities differ by orders of magnitude, we implement adaptive dynamic sensitivity scaling. At each iteration i , scaling factors $S_C^{(i)}$ and $S_\mu^{(i)}$ are evaluated as the mean absolute values (L_1 -norms) of the raw gradients:

$$S_C^{(i)} = \frac{1}{m} \sum_{e=1}^m \left| \frac{\partial C^{(i)}}{\partial \hat{x}_e} \right|, \quad S_\mu^{(i)} = \frac{1}{m} \sum_{e=1}^m \left| \frac{\partial \mu_1^{(i)}}{\partial \hat{x}_e} \right| \quad (34)$$

The aggregated sensitivity passed to the optimizer is:

$$\frac{\partial F^{(i)}}{\partial \hat{x}_e} = (1 - \alpha) \frac{1}{S_C^{(i)}} \frac{\partial C^{(i)}}{\partial \hat{x}_e} + \alpha \frac{1}{S_\mu^{(i)}} \frac{\partial \mu_1^{(i)}}{\partial \hat{x}_e} \quad (35)$$

Under Geoffrion's theorem [10], dividing by positive scaling norms mathematically guarantees convergence to a properly efficient Pareto-optimal solution.

5 Accelerated Heaviside Continuation for Slender 3D Topologies

To transform intermediate densities into distinct $\{0, 1\}$ structural topologies, filtered densities $\tilde{x}_e$ are mapped through a smoothed Heaviside projection [8]:

$$\hat{x}_e = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{x}_e - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \quad (36)$$

where $\eta \in (0, 1)$ is the threshold parameter (typically $\eta = 0.5$) and $\beta \in [1, \infty)$ dictates the sharpness of the transition interface.

5.1 Pathologies of Static Low β and Abrupt High β

The magnitude of β governs the regularization of the step function. When β is maintained at a low constant value ($\beta = 1$), the projection remains smooth and well-behaved, facilitating broad topological exploration. However, persistent low β produces wide “grey zones” of intermediate density ($0 < \hat{x}_e < 1$) across element boundaries. In 3D linearized buckling optimization, extended grey zones distort the differential penalization ratio $E_G(\hat{x}_e)/E_K(\hat{x}_e)$, creating non-physical boundary compliance and obscuring localized structural bifurcation modes.

Conversely, abruptly initializing or stepping to high β (e.g., $\beta \geq 16$) causes severe gradient vanishing away from the threshold interface η :

$$\frac{\partial \hat{x}_e}{\partial \tilde{x}_e} = \frac{\beta \operatorname{sech}^2(\beta(\tilde{x}_e - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \xrightarrow{|\tilde{x}_e - \eta| > \delta} 0 \quad (37)$$

This premature vanishing traps the design variables in sub-optimal local extrema, freezing chunky, stubby, and over-bulked structural members that fail to branch into slender, high-efficiency truss topologies.

5.2 Accelerated Continuation Formulation

To eliminate intermediate grey zones while simultaneously resolving slender structural features, an accelerated continuation schedule for β is enforced:

$$\beta^{(k+1)} = \min\left(\beta_{\max}, \gamma_{\beta} \cdot \beta^{(k)}\right) \quad (38)$$

where $\beta^{(0)} = 1$, $\gamma_{\beta} \in [1.5, 2.0]$ is the continuation multiplier (e.g., geometric doubling $\beta \in \{1, 2, 4, 8, 16, 32, 64\}$), and $\beta_{\max} = 32$ (or 64). The continuation step is updated according to:

$$\text{Update } \beta \quad \text{if} \quad (k \pmod{\Delta N_{\beta}} = 0) \quad \text{or} \quad \left(\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-\Delta k)}\|_{\infty} < \epsilon_{\beta}\right) \quad (39)$$

with update cadence $\Delta N_{\beta} \in [8, 12]$ iterations and convergence tolerance $\epsilon_{\beta} = 0.01$.

During the initial phase ($\beta \leq 4$), low projection steepness permits the continuous material to flow globally, establishing optimal structural connectivity between supports and loads. As the continuation accelerates β into higher regimes ($\beta \geq 8$), the transition layer narrows sharply to the element width, forcing intermediate voxels strictly to 0 or 1. This rapid boundary sharpening eliminates grey zones and prevents the accumulation of chunky, stubby material blocks, compelling the optimizer to discover slender, high-aspect-ratio tensile ties and optimized triangulated compressive bracing.

6 Sensitivity Analysis

The compliance derivative is evaluated as [7]:

$$\frac{\partial C}{\partial \hat{x}_e} = -\mathbf{u}_e^T \mathbf{k}_0 \mathbf{u}_e \frac{\partial E_K(\hat{x}_e)}{\partial \hat{x}_e} = -p_K \hat{x}_e^{p_K-1} (E_0 - E_{\min}) \mathbf{u}_e^T \mathbf{k}_0 \mathbf{u}_e \quad (40)$$

For the fundamental buckling reciprocal eigenvalue μ_1 , the exact sensitivity is derived via the adjoint state method [4, 7]:

$$\frac{\partial \mu_1}{\partial \hat{x}_e} = - \left[\phi_{1,e}^T \frac{\partial \mathbf{G}_e}{\partial \hat{x}_e} \phi_{1,e} + \mu_1 \phi_{1,e}^T \frac{\partial \mathbf{k}_e}{\partial \hat{x}_e} \phi_{1,e} - \mathbf{w}_{1,e}^T \frac{\partial \mathbf{k}_e}{\partial \hat{x}_e} \mathbf{u}_e \right] \quad (41)$$

where:

- ϕ_1 : eigenvector / fundamental buckling mode shape, normalized such that $\phi_1^T \mathbf{K} \phi_1 = 1$.
- μ_1 : fundamental reciprocal eigenvalue ($1/\lambda_1$).
- $\mathbf{u}_e$: elemental static displacement vector under external loads $\mathbf{F}$.
- $\mathbf{w}_1$: global adjoint displacement vector, obtained by solving:

$$\mathbf{K} \mathbf{w}_1 = \mathbf{F}_{\text{adj}} = \sum_{e=1}^m E_G(\hat{x}_e) \mathbf{B}_0^T \mathbf{D}^T \mathbf{P}_e \quad (42)$$

where $P_{e,k} = \phi_{1,e}^T \mathbf{G}_{0,k} \phi_{1,e}$ for $k \in \{xx, yy, zz, yz, xz, xy\}$.

Crucially, evaluating $\mathbf{w}_1$ requires **zero new matrix factorizations**: the pre-factored stiffness matrix from the forward static solve $\mathbf{K} \mathbf{U} = \mathbf{F}$ is directly reused for a single rapid back-substitution.

7 Updating Process

The physical sensitivities are back-propagated through the Heaviside projection and filter chain rule:

$$\frac{\partial \tilde{F}}{\partial x_i} = \sum_{e=1}^m \frac{H_{ei}}{\sum_k H_{ek}} \left(\frac{\partial F}{\partial \hat{x}_e} \frac{\partial \hat{x}_e}{\partial x_i} \right) \quad (43)$$

The design variables are updated using the Optimality Criteria (OC) scheme [2, 7]:

$$x_e^{(k+1)} = \begin{cases} \max(x_{\min}, x_e^{(k)} - \zeta) & \text{if } x_e^{(k)} B_e \leq \max(x_{\min}, x_e^{(k)} - \zeta) \\ x_e^{(k)} B_e & \text{if } \max(x_{\min}, x_e^{(k)} - \zeta) < x_e^{(k)} B_e < \min(1, x_e^{(k)} + \zeta) \\ \min(1, x_e^{(k)} + \zeta) & \text{if } \min(1, x_e^{(k)} + \zeta) \leq x_e^{(k)} B_e \end{cases} \quad (44)$$

where $\zeta = 0.2$ is the iteration move limit, $x_{\min} = 0.001$, and:

$$B_e = \sqrt{\max \left(0, -\frac{\partial \tilde{F} / \partial x_e}{\lambda_L \cdot \partial V / \partial x_e} \right)} \quad (45)$$

The volume Lagrange multiplier λ_L is determined via bisection search until volume conservation $|\frac{1}{m} \sum_e \hat{x}_e - V_f| < 10^{-4}$ is satisfied.

8 Post-Processing: Discrete Surface Smoothing via Taubin Volume-Preserving Filtering

Upon convergence of the filtered and projected pseudo-density field $\hat{\mathbf{x}}^*$, an explicit geometric boundary of the structural topology is extracted at the discrete isosurface $\hat{x}_{\text{iso}} = 0.5$. In the Cartesian voxel framework, this is accomplished by traversing the computational grid to detect all external boundary quad facets separating solid voxels ($\hat{x}_e \geq 0.5$) from void voxels ($\hat{x}_e < 0.5$). Each rectangular quadrilateral facet is split along its shortest diagonal into two coplanar triangles, yielding an orientable, watertight 2-manifold triangular boundary mesh:

$$\mathcal{M} = (\mathcal{V}, \mathcal{F}) \quad (46)$$

parameterized by the vertex coordinate set $\mathcal{V} = \{\mathbf{v}_i \in \mathbb{R}^3\}_{i=1}^{N_v}$ and the triangular face connectivity set $\mathcal{F} = \{f_k\}_{k=1}^{N_f}$.

Because the boundary mesh is extracted directly from orthogonal hexahedral cells, raw exported isosurfaces exhibit severe high-frequency voxel staircase artifacts. These microscopic discontinuities create spurious stress concentrations in downstream validation analyses and introduce unacceptable surface roughness for additive manufacturing and slicer toolpath generation.

8.1 Discrete Laplace-Beltrami Operator on 2-Manifold Boundary Meshes

Surface smoothing is formulated directly on the discrete graph structure of $\mathcal{M}$ through the discrete Laplace-Beltrami operator. For any boundary vertex $\mathbf{v}_i \in \mathcal{V}$, the discrete surface Laplacian $\Delta \mathbf{v}_i$ represents the vector displacement between $\mathbf{v}_i$ and the convex combination of its adjacent 1-ring neighbors:

$$\Delta \mathbf{v}_i = \sum_{j \in \mathcal{N}(i)} w_{ij} (\mathbf{v}_j - \mathbf{v}_i) \quad (47)$$

where $\mathcal{N}(i) = \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$ denotes the set of immediate topological neighbors connected to vertex i by an edge in $\mathcal{E}$, and w_{ij} are non-negative normalized edge weights satisfying the partition of unity:

$$\sum_{j \in \mathcal{N}(i)} w_{ij} = 1 \quad (48)$$

Under the discrete umbrella operator (uniform weighting), the weights are simply evaluated as:

$$w_{ij} = \frac{1}{|\mathcal{N}(i)|} \quad (49)$$

where $|\mathcal{N}(i)|$ is the valence (degree) of vertex $\mathbf{v}_i$.

8.2 Volumetric Shrinkage of Classical Laplacian Smoothing

Classical Laplacian smoothing iteratively updates vertex coordinates through the forward Euler discretization of the isotropic surface heat diffusion equation $\partial \mathbf{v} / \partial t = \Delta \mathbf{v}$:

$$\mathbf{v}_i^{(k+1)} = \mathbf{v}_i^{(k)} + \lambda \Delta \mathbf{v}_i^{(k)}, \quad \lambda > 0 \quad (50)$$

While explicit Laplacian smoothing effectively damps high-frequency spatial noise, it acts as a low-pass Gaussian filter that induces severe global volume loss and geometric shrinkage. For slender structural members—such as thin tensile ties and delicate compressive trusses—repeated Laplacian iterations steadily erode cross-sectional dimensions, substantially shrinking the structural volume and degrading the buckling load capacity.

8.3 Dual-Step Taubin Volume-Preserving Smoothing Filter

To suppress high-frequency voxel staircase noise without shrinking the macroscopic volume or rounding slender structural chords, we incorporate the non-shrinking dual-step filter formulated by Taubin [11]. Within each smoothing cycle $k \in \{0, 1, \dots, N_{\text{cycles}} - 1\}$, the algorithm alternates between an inward diffusion step and an outward anti-diffusion step:

$$\mathbf{v}_i^{(2k+1)} = \mathbf{v}_i^{(2k)} + \lambda \Delta \mathbf{v}_i^{(2k)} \quad (51)$$

$$\mathbf{v}_i^{(2k+2)} = \mathbf{v}_i^{(2k+1)} + \mu \Delta \mathbf{v}_i^{(2k+1)} \quad (52)$$

with positive shrink parameter $\lambda > 0$ and negative unshrink parameter $\mu < -\lambda < 0$, satisfying the volume-preservation inequality:

$$\frac{1}{\lambda} + \frac{1}{\mu} < 0 \quad (53)$$

In the spectral domain of the discrete Laplace-Beltrami operator, with spatial frequency eigenvalues $k \in [0, k_{\text{max}}]$ (where $k_{\text{max}} \leq 2$ on normalized meshes), the transfer function over a complete two-step iteration cycle is:

$$f(k) = (1 - \lambda k)(1 - \mu k) = 1 - (\lambda + \mu)k + \lambda \mu k^2 \quad (54)$$

The mathematical condition $\mu < -\lambda < 0$ and $\frac{1}{\lambda} + \frac{1}{\mu} < 0$ guarantees that:

1. **Zero Low-Frequency Shrinkage:** At low spatial frequencies ($k \rightarrow 0$), the transfer function maintains unit gain ($f(0) = 1$) with positive slope $f'(0) = -(\lambda + \mu) > 0$. The anti-diffusive step ($\mu < 0$) counteracts the shrinkage induced by the positive diffusion step ($\lambda > 0$), perfectly stabilizing the global enclosed volume.
2. **Attenuation of High-Frequency Staircase Artifacts:** For high spatial frequencies $k \in (k_{\text{pass}}, 2]$, the magnitude of the transfer function satisfies $|f(k)| < 1$, ensuring monotonic damping of the Nyquist-frequency staircase artifacts associated with voxel edge discretization.

In the production pipeline, stable filter parameters $\lambda = 0.50$ and $\mu = -0.53$ are applied over $N_{\text{cycles}} = 10$ – 20 iterations. This produces smooth, curvature-continuous, and volume-preserving 2-manifold triangular meshes that strictly preserve the mechanical stiffness, cross-sectional area, and Euler buckling resistance of slender 3D structural topologies.

9 Numerical Performance on 8GB Hardware

To guarantee execution on standard laptop hardware (AMD Ryzen 7 4700U with 8GB RAM), the solver employs:

1. **Preconditioned Conjugate Gradient (PCG):** Jacobi diagonal preconditioning coupled with displacement warm-starting achieves convergence in $O(\sqrt{\kappa})$ iterations while using only ≈ 12 KB RAM per element (reducing solver memory by $18\times$ compared to direct LU factorizations).
2. **Invariant Assembly Indexing:** Sparse row and column coordinate vectors (iK, jK) are pre-computed once and shared between elastic stiffness $\mathbf{K}$ and geometric stiffness $\mathbf{G}$.
3. **Watertight STL Export:** A boundary-face extraction algorithm identifies only external quad facets between solid ($\hat{x}_e \geq \text{threshold}$) and void voxels, producing closed, manifold triangular meshes without internal partition planes, directly consumable by additive manufacturing slicers and PicoGK OpenVDB pipelines.

Table 1: Memory scaling and performance across 3D voxel grid resolutions (Ryzen 7 4700U, 8GB RAM).

Grid ($N_x \times N_y \times N_z$)	Elements	Total DOFs	Sparse RAM	Peak RAM	Suitability
$10 \times 10 \times 10$	1,000	3,993	~ 7 MB	$< \mathbf{60}$ MB	Fast Unit Test
$20 \times 10 \times 10$	2,000	7,623	~ 14 MB	$< \mathbf{120}$ MB	Interactive Dashboard
$30 \times 15 \times 10$	4,500	16,368	~ 31 MB	$< \mathbf{310}$ MB	High-Fidelity Domain
$40 \times 20 \times 10$	8,000	28,413	~ 55 MB	$< \mathbf{620}$ MB	Production Domain

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